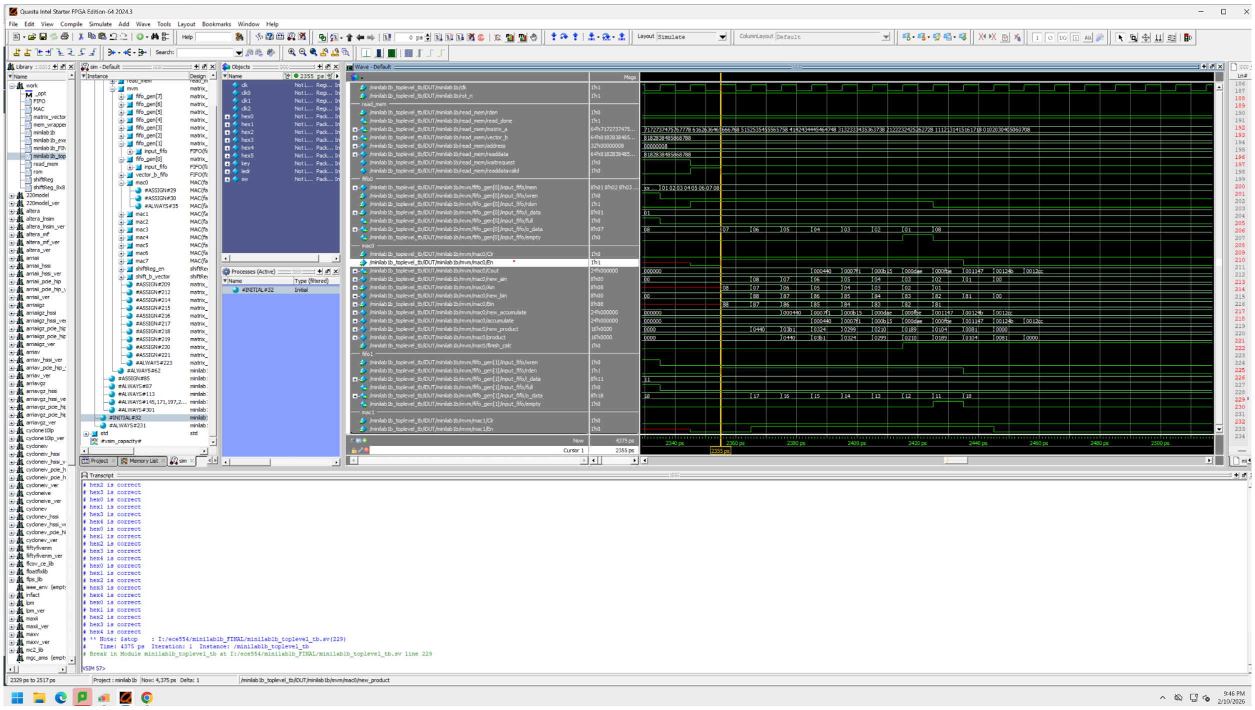
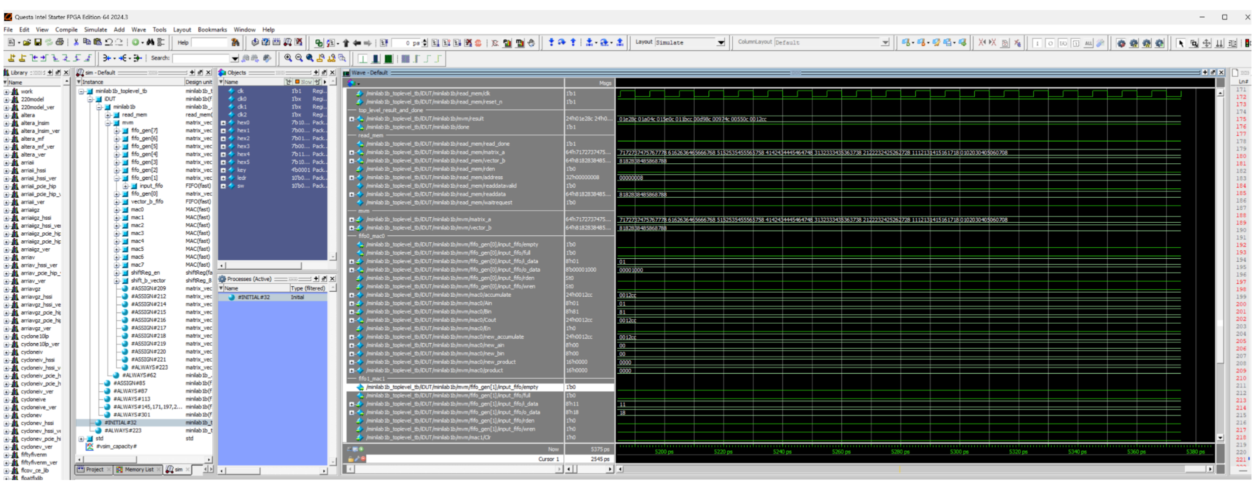
How you tested the design via simulation (screenshot of waveform).
To test the design I created a test bench that ran based off my created matrix and vector of numbers. From there I looked at the waveform mainly following the enable signals to ensure that I was getting the expected value and result.

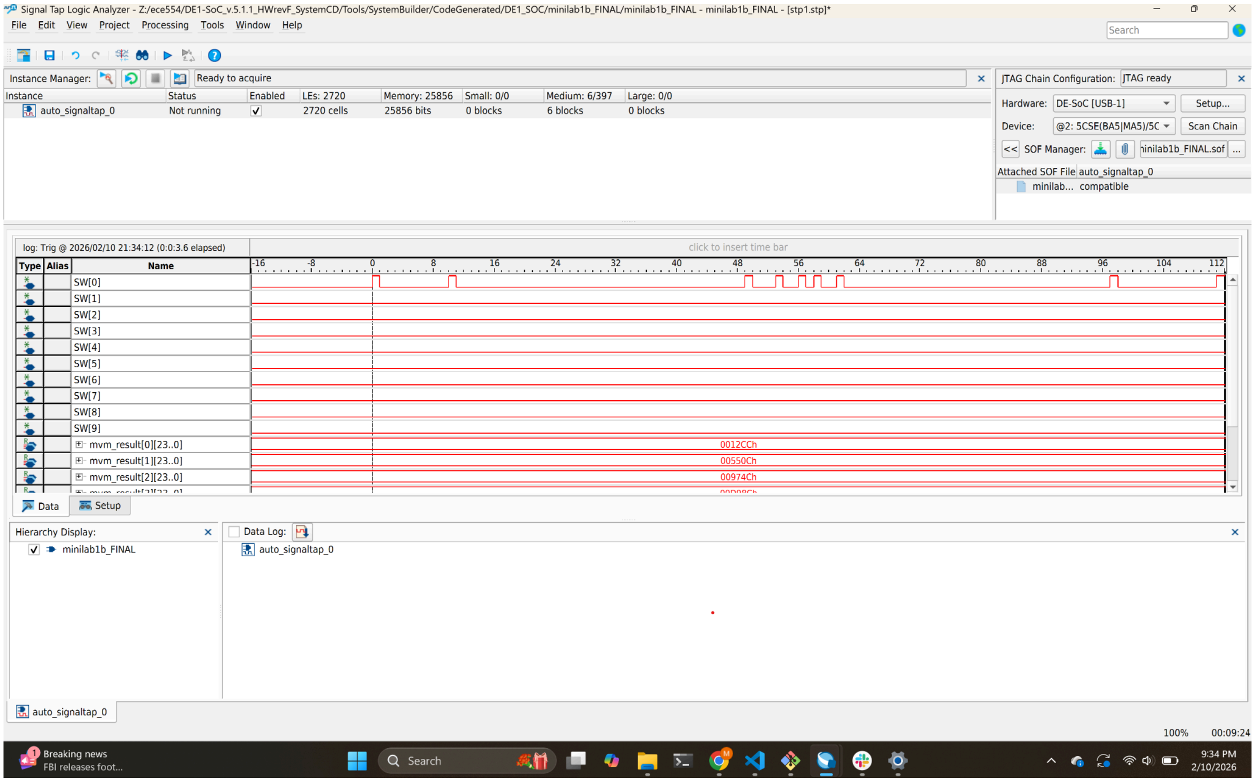




How you fixed the design to meet the timing constraint.
To fix the timing constraint I had to add flops between my fifo data coming out before inputting into the mac. That helped the slack but still wasn't the full fix. I then had to add flops in the mac so that it would take the output of the product flop it into the adder since both of these operations took a lot of time.

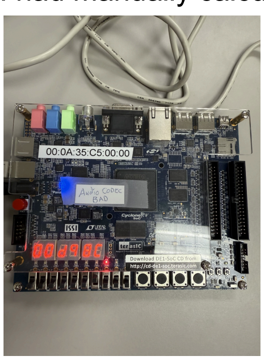
How you tested the design using SignalTap (screenshot of waveform around the trigger condition).

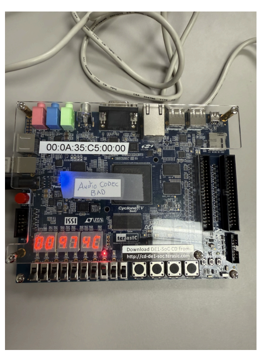
For a small bit I was seeing an incorrect value with the switches due to a clocking issue. I used the switches as a trigger and watched the results along with some other done signals.

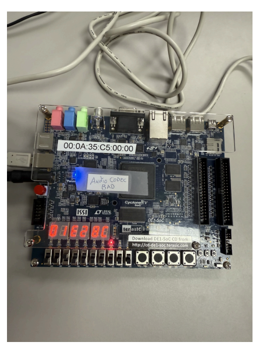


How you tested the design on board using an example and snapshot of the board that shows it.

On the board I flipped each switch to ensure I was seeing the correct hex values on the screen. I had manually calculated the values from the memory to do this check.







Explain any difficulties that you faced during the whole process and how you overcame it.

When I was trying to fix timing I kept getting incorrect values due to the new flopped data. I had to draw out tables with the new scenarios showing when the correct data was available and had to add a couple of new enables so that it was still calculating after the enables were done but not taking any new values. This went back and forth a couple of times where I would fix timing but break the functionality but eventually got both to work. It took a lot of simulation and waveforms along with some diagrams.